

1. What is Computer Vision?

Computer Vision is a field within Artificial Intelligence that empowers machines, such as cameras and computers, to interpret and derive meaningful information from visual data like digital images and videos. It aims to enable machines to achieve a high-level understanding of visual inputs, allowing them to identify objects, scenes, and actions, and subsequently make informed decisions or take appropriate actions based on this understanding. The general process involves acquiring images, pre-processing them, analyzing their content, and finally extracting and interpreting the relevant information.

2. What are pixels and how do they represent an image?

An image is fundamentally composed of a grid of small, square elements known as pixels. Each pixel in an image holds its own color information, and the entire image is simply a combination of these individual colored pixels arranged in a two-dimensional grid defined by the image's height and width (its size, measured in the number of pixels). Each pixel also has an intensity value, typically ranging from 0 to 255, which determines the brightness of that pixel.

3. What are color spaces and color channels, and how do they define color in an image?

A color space is a specific organization of colors. Common examples include Grayscale, RGB (Red, Green, Blue), and HSV (Hue, Saturation, Value). Color channels are the individual components that make up a color space. For instance, the RGB color space has three channels: Red, Green, and Blue. In an RGB image, the color of each pixel is determined by the intensity values (ranging from 0 to 255) in each of these three channels. A value of 0 indicates the absence of that color component, while 255 signifies its maximum intensity. Combining different intensities of red, green, and blue allows for a vast spectrum of colors to be represented. Grayscale images, on the other hand, have only one channel where each pixel's intensity value represents a shade of gray, ranging from black (0) to white (255).

4. Why is grayscaling important in computer vision?

Grayscaling, the process of converting a color image to a grayscale image, is significant in computer vision primarily because it reduces the dimensionality of the image data. An RGB image has three color channels, whereas a grayscale image has only one. This reduction in data simplifies computations, lowers the computational cost for algorithms, and can improve processing speed. Furthermore, for many computer vision tasks, the essential structural and shape information of objects is often preserved in the



In terms of pixel intensity values, edges in an image are regions where there is a sudden and significant change in brightness between neighboring pixels. Mathematically, this is represented by a large gradient or shift in pixel intensity values in a particular direction. Edges are crucial because they often contain the most important information about the shapes and boundaries of objects within an image. Object recognition and detection can frequently be performed effectively by focusing on the edges, even with minimal background or detailed information present.

6. How does convolution with a kernel help in detecting edges in an image?

Edge detection can be achieved by performing a convolution operation between an input image (represented as a pixel map) and a small array of numbers called a filter or kernel. This kernel is designed with a specific pattern of numbers that, when convolved with the image, will produce an output feature map where edges corresponding to that pattern are enhanced. The convolution process involves sliding the kernel over the image, computing the sum of the element-wise products between the kernel and the underlying image patch, and placing this sum into the corresponding location in the output feature map. Kernels with specific patterns, like the vertical edge detector (e.g., with values like -1, 0, 1 arranged vertically), will yield high output values in regions of the image where the pixel intensity changes sharply in the horizontal direction (indicating a vertical edge).

7. What are some common edge detection filters (kernels) and what are their characteristics?

Several popular convolution filters are used for edge detection, including the Prewitt, Sobel, and Laplacian filters. The Prewitt filter has variations for detecting horizontal and vertical edges and its elements often sum to zero with opposite signs. The Sobel filter is similar to Prewitt but assigns more weight to the central pixels in the kernel (e.g., -2, 0, 2 instead of -1, 0, 1), often resulting in stronger edge responses. The Laplacian filter is a second-order derivative filter that detects regions of rapid intensity change and can detect edges in all directions with a single kernel, although it is also more sensitive to noise.

